

SolarGuard AI

AI-Powered Microcrack Detection for Solar Farms

Investor Pitch Deck · 2026

\$200K seed · Profitable from Year 1 · NPV +\$4M

The Problem: Silent Revenue Killer



Microcracks start invisibly

Initiated during manufacturing or thermal cycling — undetectable to the eye until too late



Progressive S-curve degradation

Loss accelerates mid-life; up to 20% capacity lost over 25 years



USD 5.93M hidden loss per farm

Al Kharsaah 800 MWp loses USD 5.93M over 25 years (base case $r=0.254$)



Current solutions fail

Annual visual inspection & IV-curve testing miss early-stage microcracks entirely

SolarGuard AI: The Solution

1

Off-the-Shelf Sensors

Commercial EL cameras + Jetson edge compute — no custom hardware required. Deploy in days.

2

Proprietary AI Model

Fine-tuned CNN detects crack signatures weeks before power loss begins. Runs on-device.

3

Fleet Dashboard

Real-time alerts, degradation curves, and maintenance priority queue via web & mobile.

4

ROI Reports

Monthly automated reports quantify MWh recovered vs. SolarGuard subscription cost.

Why \$200K is Enough to Build This

Off-the-shelf hardware eliminates custom R&D. A 2-person remote team handles everything.

Component	Lean Approach	Cost
AI Model	Fine-tune open-source CNN (PyTorch)	~\$0 (cloud GPUs) ~\$5K/yr
EL Camera	Commercial Solibot or similar	\$5K–15K per unit
Edge Compute	NVIDIA Jetson Nano	\$500–1K per site
Dashboard	React + AWS Serverless	\$10K dev + \$500/mo hosting
Team (Yr 1)	Founder + 1 remote AI engineer	\$130K/yr total
Pilot Hardware	Full kit for 1 farm	\$25K–30K
Legal + Misc	IP filing, incorporation	\$20K

Total first-year burn: ~\$180K — leaving buffer within \$200K seed

Business Model

SaaS Subscription

USD 47,500/yr

Annual license per farm
Customer keeps 80% of savings

Hardware + Setup

USD 400,000

One-time · 25% COGS (off-the-shelf)
USD 250K HW + USD 150K impl.

Metric	Value
SaaS gross margin	70% (30% COGS)
HW + impl gross margin	75% (25% COGS, off-the-shelf)
Customer 25-yr SaaS cost	USD 1.19M
Customer 25-yr loss prevented	USD 5.93M (base case)
Customer ROI	~250% on SolarGuard cost

Financial Projections

Year	Farms	SaaS ARR (\$M)	Total Rev (\$M)	EBITDA (\$M)	Margin
1	1	0.05	0.45	0.20	45.4%
2	2	0.10	0.49	0.20	39.7%
3	3	0.14	0.54	0.22	40.5%
5	8	0.38	1.58	0.97	61.1%
7	17	0.81	2.81	1.82	64.7%
10	30	1.43	3.03	1.95	64.4%

Seed Required

\$200K

Lean startup

Profitable

Year 1

EBITDA +\$203K

10-yr Revenue

\$18.0M

Total

NPV @ 15%

+\$4.0M

Positive!

Customer ROI: 250% Return on Investment

For every \$1 spent on SolarGuard AI, the customer gets \$3.50 back.

Total 25-yr Loss: USD 5.93M

Recoverable (70%): USD 4.15M

SolarGuard 25-yr SaaS cost: USD 1.19M (20% of loss)

Customer invests

USD 1.19M

25-yr SaaS fee

Customer recovers

USD 4.15M

70% of loss prevented

Customer net gain

~USD 3.0M

Over 25 years

Customer ROI

~250%

On SolarGuard cost

Capital Efficiency: \$200K Seed Breakdown

50% \$100K — AI/ML fine-tuning
+ cloud compute

15% \$30K — Pilot hardware kit
(EL camera + Jetson)

20% \$40K — Operations & sales
(remote team)

10% \$20K — Legal / IP / registration

5% \$10K — Contingency

\$200K · Profitable from Year 1 · No further dilution needed until Series A at Year 5

Go-to-Market: 30 Farms by Year 10

Phase 1 Yr 1–2

3 farms

Qatar pilot

- ▶ Al Kharsaah anchor client
- ▶ Deploy off-the-shelf kit
- ▶ Collect data, tune model

Phase 2 Yr 3–5

8 farms

MENA expansion

- ▶ Saudi & UAE solar farms
- ▶ Channel partner deals
- ▶ Self-serve SaaS onboarding

Phase 3 Yr 6–10

30 farms

Scale to SOM

- ▶ Expand MENA + Africa
- ▶ Remote hub model
- ▶ \$1.43M SaaS ARR

Protect Every Photon.

\$200K seed. Profitable Year 1. No complex hardware.
Just AI + off-the-shelf sensors + a compelling ROI.

Seeking: USD 200K Seed · 30 Farms by Year 10 · NPV +\$4M

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